

ADITYA PRADEEP KOGRI

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CAREER SUMMARY

Graduate Electrical Engineering student at Arizona State University with hands-on experience in design verification, RTL design, logic synthesis, and physical design across 7nm FinFET and 32nm process technologies. Built a **UVM testbench** from scratch with constrained-random stimulus, functional coverage, and SystemVerilog Assertions, and completed a **full ASIC flow from RTL through GDSII signoff**. Proficient in SystemVerilog, Verilog, TCL, and C++ with scripting experience for design analysis and regression runs. Seeking **full-time ASIC Design and Verification** roles starting December 2026.

EDUCATION

Master's Degree | Electrical Engineering

Arizona State University, Tempe, AZ

Graduating December 2026

GPA: 3.42

Coursework: VLSI Design, ASIC Design Flow, Digital Systems and Circuits, Verilog HDL, Digital Verification and Testing, Python

Bachelor's Degree | Electrical and Electronics Engineering

University Visveswaraya College of Engineering, Bengaluru, India

August 2024

GPA: 8.11/10

Coursework: Digital Signal Processing, Analog Integrated Circuits, CMOS Transistors, Microcontrollers

TECHNICAL SKILLS

Verification: UVM, SystemVerilog Assertions (SVA), Constrained-Random Verification, Functional Coverage, Scoreboarding, Testbench Development

ASIC/VLSI Design: RTL Design, Logic Synthesis, Timing Analysis, Physical Design (Floorplanning, Placement, CTS, Routing), DRC/LVS

EDA Tools: Cadence Xcelium, Innovus, Virtuoso, Synopsys Design Compiler, QuestaSim, ModelSim, HSPICE, LTspice

Programming & Scripting: SystemVerilog, Verilog, TCL, C++, Python (Basic), MATLAB, SQL (Basic), Git

Operating Systems: Linux, Windows

ACADEMIC PROJECTS

Synchronous FIFO - UVM Verification (SystemVerilog, Cadence Xcelium)

Fall 2026

- Built a **complete UVM testbench** (driver, monitor, scoreboard, agent/env/test hierarchy) for a parameterized 8x8 synchronous FIFO, running directed and **constrained-random** sequences side by side.
- Random tests alone stopped at 83% coverage since the FIFO never hit full, so I wrote a targeted fill sequence to close the gap and reached **100% functional coverage** with 0 scoreboard errors.
- Added **SVA checks bound to the DUT** for full/empty and overflow/underflow, and **root-caused 4 race-condition bugs** (reset, driver/DUT, monitor sampling) by tracing waveforms.

ASIC Acceleration for GCN (Graph Convolution Neural Network) - RTL to GDSII

Fall 2025

Tools: Synopsys Design Compiler, Cadence Innovus, ModelSim/QuestaSim (7nm PDK)

- Developed testbenches and ran functional verification of convolution and pooling modules in ModelSim, validating correctness across multiple edge-case scenarios before synthesis
- Ran logic synthesis in Synopsys Design Compiler under timing, area, and power constraints analyzed trade-offs across design iterations to meet target performance
- Drove complete physical design flow in Cadence Innovus (floorplanning, placement, CTS, routing) and achieved clean DRC/LVS signoff, ensuring manufacturing quality

4-Bit Adder Design and Implementation (SAED 32nm, Cadence Virtuoso)

Spring 2025

- Designed schematic and **transistor-level implementation** of a 4-bit mirror adder in **Cadence Virtuoso**, integrating 1-bit adder cells and inverters.
- Analyzed **worst-case delay, average power, and layout area** to evaluate performance and identify design bottlenecks.
- Achieved **DRC/LVS clean** physical layout, demonstrating end-to-end closure from schematic through physical verification.

Delay Optimization of Logic Path (32nm PDK)

Spring 2025

- Designed a static CMOS logic path using INVx01, NAND3x03, NOR2x05 standard cells and applied logical techniques to minimize critical path delay
- Analyzed timing reports and parasitic extraction results to evaluate delay, power, and layout-dependent effects
- Wrote scripted automation to compare performance across sizing iterations and validate optimization decisions against design targets

PROFESSIONAL EXPERIENCE

Teleindia Networks Pvt Ltd GET Trainee - Cloud (IT Team)

July 2024 – December 2024

- Managed and configured Windows and **Linux virtual machines**, building proficiency in Linux-based environments used in ASIC design workflows.
- Deployed, tested, and maintained web applications on cloud-based servers, gaining exposure to **infrastructure management** and debugging practices.
- Collaborated cross-functionally with IT teams to troubleshoot and document technical issues**, building communication habits directly applicable to cross-team engineering work